Path A3: Local Caching & Cloud Sync Architecture for Decay State Bookmarks

1. Architectural Overview & Sync Lifecycle

The **Decay State Bookmark Engine** bridges local device playback states with the Netflix cloud infrastructure (EVCache transient store & Cassandra viewing history). It guarantees that even if a user falls asleep in an offline state (e.g., streaming downloaded content on a flight or experiencing overnight Wi-Fi disconnects), their exact **Decay Bookmark Timestamp (TS_{decay})** is stored locally first, then synchronized reliably across all profile-linked devices upon network reconnection.



2. Multi-Platform Local Caching Layer

Each client platform implements an **Offline-First Storage Adapter** using native, zero-latency local database engines to write decay bookmarks instantly at the start of the display dimming transition curve (T_{-10m}).

Target Platform	Local Engine	Encryption & Security	Write Throughput Guarantee

iOS / iPadOS	CoreData + SQLite	Data Protection API (NSFileProtectionCompleteUnlessOpen)	$< 5ms$ synchronous local write
Android / Google TV	Room Persistence Library	EncryptedSharedPreferences + SQLCipher	$< 5ms$ non-blocking coroutine write
Smart TV OS (WebOS, Tizen, FireTV)	IndexedDB / Native SQLite	OS Secure Storage Sandbox	$< 12ms$ async transactional write

Local Cache Schema (decay_bookmarks_v1)

```
SQL
CREATE TABLE decay_bookmarks_v1 (
    bookmark_id TEXT PRIMARY KEY NOT NULL,
    account_id TEXT NOT NULL,
    profile_id TEXT NOT NULL,
    show_id TEXT NOT NULL,
    episode_id TEXT NOT NULL,
    decay_timestamp_ms INTEGER NOT NULL,
    fade_start_timestamp_ms INTEGER NOT NULL,
    trigger_type TEXT NOT NULL, -- 'manual', 'schedule', 'wearable_auto'
    sync_status TEXT DEFAULT 'PENDING_SYNC', -- 'PENDING_SYNC', 'SYNCED', 'CONFLICT'
    device_id TEXT NOT NULL,
    created_at_utc INTEGER NOT NULL,
    updated_at_utc INTEGER NOT NULL
);

CREATE INDEX idx_profile_sync ON decay_bookmarks(profile_id, sync_status);
```

3. Distributed Sync & Conflict Resolution Engine

Conflict Resolution Strategy: Last-Write-Wins CRDT

(LWW-Element-Set)

When multiple devices under the same profile generate conflicting state updates (e.g., an iPad goes offline while a TV continues streaming under the same profile), the sync engine uses a **Vector Clock + Logical Timestamp CRDT** model to resolve conflicts without data loss.



Conflict Resolution Matrix

Scenario	Local State	Remote Cloud State	Resolved Outcome	Rationale
Offline Recovery	Offline Decay BM: 14:20	No Cloud Update	**Offline Local BM Prevails**	App renders recovery card directly from local cache upon next launch.
Dual Device Profile Conflict	TV Decay BM: 14:20	Phone Progress BM: 42:10	**Decay BM 14:20 Wins**	Sleep markers override standard

				auto-play progress to protect from spoilers.
Manual Override	Decay BM: 14:20	User manually clicks "Scrub to 45:00"	**Manual Scrub 45:00 Wins**	Active human user input invalidates passive decay markers.

4. Cloud Sync Network Protocol & Payload

The network sync manager leverages **Background Synchronization APIs** (iOS BGAppRefreshTask, Android WorkManager, Web Service Workers) to push pending records when network connectivity is restored.

Client Sync Dispatch Request (POST /v1/telemetry/sleep/sync)

JSON


```
{
  "sync_header": {
    "client_tx_id": "tx_99812039-44ab",
    "device_id": "dev_ios_mac_8839",
    "network_type": "WIFI_RESTORED",
    "retry_count": 0
  },
  "pending_bookmarks": [
    {
      "bookmark_id": "bm_8839201_80192011",
      "account_id": "usr_8839201",
      "profile_id": "prf_4401",
      "show_id": "80057281",
      "episode_id": "80192011",
      "decay_timestamp_ms": 860400,
      "fade_start_timestamp_ms": 740400,
      "trigger_type": "wearable_auto",
      "device_timestamp_utc": 1776483735000
```

```
    "vector_clock": {
      "device_node": "dev_ios_mac_8839",
      "counter": 4
    }
  }
}
```

Server Sync Ack & Recovery Card Payload (200 OK)

```JSON
{
  "sync_response": {
    "status": "SUCCESS",
    "processed_count": 1,
    "conflict_detected": false
  },
  "recovery_card_state": {
    "render_card": true,
    "profile_id": "prf_4401",
    "headline": "You fell asleep during Episode 3",
    "subtext": "Faded out at 14:20 (decay point preserved)",
    "resume_action": {
      "show_id": "80057281",
      "episode_id": "80192011",
      "seek_target_ms": 860400
    }
  }
}
```

5. Offline Resiliency & Fallback Verification Matrix

Network Condition	System Reaction	User Experience
Instant Offline (Flight / Tunnel)	Write directly to local SQLite/Room database with sync_status = 'PENDING_SYNC'.	App functions offline seamlessly; morning recovery card loads from local DB.

Intermittent / Flaky Wi-Fi	Retries exponential backoff $(2s, 4s, 8s, 16s)$ via background sync worker.	No UI blocking or network timeout alerts displayed to the sleeping user.
App Forced Closed During Fade	Persistent transaction log flushes decay timestamp immediately at T_{-10m}.	Opening Netflix after force-quit still presents the exact decay recovery timestamp.